

No-Code Automation with n8n

Enterprise Orchestration Layer

Practical Training Module for Insurance, Marketing and Sales Automation Teams

Program Objective

Build enterprise automation workflows without heavy coding using n8n. Participants will design, test, and deliver an end-to-end lead-processing workflow that captures leads, classifies them using AI, pushes records to a CRM-style system, and sends an automated email response.

1. Overview

Duration	4 Hours / Half Day
Audience	Beginner to Intermediate UiPath Developers, Automation Developers, Business Analysts, CRM Operations Teams
Tool Focus	n8n, Webhook Trigger, Schedule Trigger, Nodes, HTTP Request, AI Classification, CRM Integration, Email Automation
Business Scenario	Automated marketing lead qualification pipeline
Final Deliverable	End-to-end n8n automation workflow
Training Style	20% Concepts, 60% Hands-on Build, 20% Review and Discussion

2. Learning Outcomes

- Understand n8n as a no-code and low-code enterprise automation platform.
- Explain the role of triggers, nodes, connections, credentials, executions and workflow design.
- Create a webhook-based lead capture workflow.

- Use AI to classify and prioritize leads.
- Push qualified lead data to a CRM-style system using HTTP Request or spreadsheet/database nodes.
- Send automated email responses to leads.
- Add error handling, data validation and basic governance checkpoints.
- Deliver an end-to-end marketing lead qualification automation workflow.

3. Session Plan

Time	Topic	Activity
20 min	n8n architecture	Concept walkthrough with enterprise use cases
25 min	Triggers	Webhook, schedule and email trigger examples
25 min	Nodes and workflow design	Design a lead processing workflow map
30 min	API integration without code	HTTP Request and CRM-style API example
35 min	AI nodes and classification	Use AI to classify leads
45 min	Hands-on workflow build	Lead capture to AI classification
35 min	CRM and email automation	Push to CRM and send response
25 min	Testing and troubleshooting	Execution review, error handling and logs

4. Concept: What is n8n?

n8n is a workflow automation platform used to connect apps, APIs, databases, email systems, AI services and business tools. It allows teams to automate processes visually using nodes, while still allowing code when needed.

In enterprise automation, n8n works as an orchestration layer. It receives events, transforms data, calls APIs, sends notifications, invokes AI services and routes outputs to downstream systems.

Simple Explanation

Think of n8n as a visual control room. Each node performs one task. The workflow connects tasks into a business process. If UiPath is the robot worker, n8n can be the coordinator that routes tasks across systems.

5. Concept: n8n Architecture

A typical n8n workflow contains triggers, processing nodes, integration nodes, AI nodes and output nodes.

Component	Meaning	Insurance / Marketing Example
Workflow	Complete automation process	Lead qualification pipeline
Trigger	Starts the workflow	New form submission from insurance website
Node	Performs one action	Send lead data to AI model
Credential	Stores secure access details	CRM API key, email SMTP credential
Execution	One workflow run	One lead submission processed
Expression	Dynamic data mapping	Use lead email from previous node

6. Concept: Triggers

A trigger starts the workflow. In n8n, common triggers include Webhook Trigger, Schedule Trigger and application-specific triggers such as email or CRM events.

Official n8n documentation describes Webhook and Schedule Trigger nodes as built-in trigger options for starting workflows.

- Webhook Trigger: Starts when another system sends HTTP data to n8n.
- Schedule Trigger: Starts at fixed times, such as every morning or every hour.
- Email Trigger: Starts when a new email arrives in a configured inbox.
- App Trigger: Starts when an event happens in a connected tool such as CRM, form app or database.

7. Trigger Examples

Trigger Type	When to Use	Lead Processing Example
Webhook	Real-time submissions	Lead form submits name, email, company and requirement
Schedule	Batch processing	Every morning, process all pending leads from spreadsheet
Email	Inbox-driven workflow	New inquiry email from prospect triggers classification

Trainer Humour

A webhook is like a doorbell. Someone submits a form, n8n says: "Someone is at the door with data."

8. Concept: Nodes and Workflow Design

A node is a building block in n8n. Each node performs one task such as reading form data, formatting fields, calling an API, sending an email or writing to a CRM.

- Trigger node starts the workflow.
- Set or Edit Fields node prepares clean data.
- IF node adds decision logic.
- HTTP Request node connects with external APIs.
- AI node or OpenAI node classifies, summarizes or drafts text.
- Email node sends communication.
- CRM or database node stores the final output.

9. Workflow Design Pattern

```
Lead Form/Webhook
-> Clean and Validate Lead Data
-> AI Lead Classification
-> IF: Hot / Warm / Cold
-> Push to CRM
-> Send Email Response
-> Notify Sales Team
-> Log Execution
```

10. Concept: API Integration Without Code

n8n can call external APIs using the HTTP Request node. This is useful when there is no ready-made CRM node or when the enterprise system exposes REST APIs.

For example, a CRM API may expect a POST request with lead data in JSON format. n8n can map fields from the webhook into the HTTP body without writing Python code.

Sample CRM API Payload

```
{
  "lead_name": "Anita Mehta",
  "email": "anita@example.com",
  "company": "ABC Insurance Brokers",
  "interest": "AI automation for claims",
  "classification": "Hot Lead",
  "priority": "High"
}
```

11. Concept: AI Nodes for Automation Enhancement

AI nodes enhance automation by understanding unstructured lead notes, emails and requirements. In a marketing pipeline, AI can classify leads, extract urgency, summarize requirements and draft personalized replies.

- Classify lead as Hot, Warm or Cold.
- Extract product interest such as claims automation, policy assistant or underwriting AI.
- Identify urgency from text.
- Generate professional email response.
- Create sales summary for CRM notes.

- Recommend next best action.

12. AI Classification Prompt Example

You are an insurance automation sales assistant.

Classify the lead as Hot, Warm or Cold.

Also return priority, insurance use case, recommended action and email response tone.

Lead Data:

Name: {{\$json.name}}

Company: {{\$json.company}}

Requirement: {{\$json.requirement}}

Budget: {{\$json.budget}}

Timeline: {{\$json.timeline}}

Return valid JSON only.

13. Case Study: Automated Marketing Lead Qualification Pipeline

Business Scenario: An insurance technology team receives leads from a website form. Manual qualification takes time and may miss urgent opportunities. The organization wants n8n to capture leads, classify them using AI, push qualified data to CRM and send automated responses.

Business Goals

- Reduce manual lead qualification effort.
- Improve response time for high-priority leads.
- Ensure consistent CRM data quality.
- Send immediate professional acknowledgement email.
- Notify sales team for hot leads.
- Create audit trail for lead processing.

14. Hands-on Lab Overview

Step	n8n Component	Outcome
1	Webhook Trigger	Capture lead from form
2	Edit Fields / Set Node	Standardize lead data
3	AI Node / OpenAI	Classify lead and generate insights

4	IF Node	Route Hot, Warm and Cold leads
5	HTTP Request / CRM Node	Push lead to CRM system
6	Email Node	Send automated email response
7	Execution Log	Review and troubleshoot workflow

15. Lab Setup Requirements

- n8n Cloud account or local n8n instance.
- OpenAI/OpenRouter API credential or organization-approved AI credential.
- SMTP or Gmail/Outlook credential for sending emails.
- CRM test endpoint, Google Sheet, Airtable or mock webhook receiver.
- Sample lead form data provided in this document.
- Browser access to n8n workflow editor.

16. Sample Lead Data for Lab

```
{
  "name": "Anita Mehta",
  "email": "anita.mehta@example.com",
  "phone": "+91-9876543210",
  "company": "ABC Insurance Brokers",
  "role": "Head of Operations",
  "requirement": "We want AI automation for claims triage and policy FAQ support.",
  "budget": "High",
  "timeline": "Within 30 days",
  "source": "Website Form"
}
```

17. Hands-on Lab 1: Create Webhook Trigger

Objective: Capture lead data from a form or test API client.

1. Open n8n and create a new workflow.
2. Add a Webhook Trigger node.
3. Set HTTP Method to POST.
4. Set Path to insurance-lead-capture.
5. Copy the Test Webhook URL.

6. Use Postman, browser extension or cURL to send sample lead JSON.
7. Confirm that n8n receives the lead data.

Expected Result

The Webhook node should show input data containing name, email, company, requirement, budget and timeline.

Practice Exercise 1

Create two more sample leads: one with urgent claims automation requirement and another with low-priority inquiry. Test both through the webhook.

18. Hands-on Lab 2: Clean and Standardize Lead Data

Objective: Prepare data for AI classification and CRM insertion.

8. Add an Edit Fields or Set node after the Webhook Trigger.
9. Create clean fields: lead_name, lead_email, company, use_case, budget, timeline and source.
10. Map each field from the incoming webhook JSON.
11. Add a new field named received_at using current timestamp if available.
12. Execute the node and verify clean output.

Example Field Mapping

Clean Field	Source Field
lead_name	{{\$.name}}
lead_email	{{\$.email}}
company	{{\$.company}}
use_case	{{\$.requirement}}
budget	{{\$.budget}}
timeline	{{\$.timeline}}

Mini Assignment 1

Add a field named lead_region. If the phone number begins with +91, set region as India. Otherwise, mark as Global.

19. Hands-on Lab 3: AI Lead Classification

Objective: Send lead details to AI and classify the lead as Hot, Warm or Cold.

13. Add an AI/OpenAI node or HTTP Request node configured to call the approved AI API.

14. Use the cleaned lead fields as prompt input.
15. Ask the model to return valid JSON only.
16. Capture fields such as classification, priority, use_case, recommended_action and email_tone.
17. Execute the node and inspect the AI output.

Recommended AI Output Schema

```
{
  "classification": "Hot Lead",
  "priority": "High",
  "insurance_use_case": "Claims triage and policy FAQ automation",
  "reason": "High budget and 30-day timeline",
  "recommended_action": "Schedule discovery call within 24 hours",
  "email_tone": "Professional and consultative"
}
```

Mini Assignment 2

Modify the AI prompt to also extract one of these solution areas: Claims Automation, Policy Assistant, Underwriting Support, Customer Support, Fraud Review, Other.

20. Hands-on Lab 4: Add IF Logic for Routing

Objective: Route leads based on AI classification.

18. Add an IF node after the AI classification node.
19. Condition: classification equals Hot Lead.
20. Route true branch to CRM push and sales notification.
21. Route false branch to CRM push and nurturing email.
22. Test Hot, Warm and Cold lead scenarios.

Routing Logic

Classification	Action	Email Type
Hot Lead	Notify sales and create CRM opportunity	Discovery call email
Warm Lead	Add to CRM and nurturing sequence	Consultative information email
Cold Lead	Add to CRM as low priority	Generic acknowledgement email

21. Hands-on Lab 5: Push Lead to CRM System

Objective: Store lead data in a CRM system or CRM-like destination. If no CRM is available, use Google Sheets, Airtable, Notion database or a mock API endpoint.

23. Add HTTP Request node or CRM node.
24. Set method to POST for API-based CRM integration.
25. Configure authentication using credentials, not hard-coded secrets.
26. Map cleaned lead data and AI classification fields into CRM payload.
27. Execute the node and verify that the lead is created.

CRM Payload Example

```
{
  "name": "{{${json.lead_name}}",
  "email": "{{${json.lead_email}}",
  "company": "{{${json.company}}",
  "source": "Website Form",
  "classification": "{{${json.classification}}",
  "priority": "{{${json.priority}}",
  "recommended_action": "{{${json.recommended_action}}"}
}
```

Mini Assignment 3

Add a CRM field named `next_follow_up_date`. For Hot Lead, set it to next working day. For Warm Lead, set it to 3 days later. For Cold Lead, set it to 7 days later.

22. Hands-on Lab 6: Send Automated Email Response

Objective: Send a professional automated response to the lead.

28. Add Email node after CRM push.
29. Configure SMTP, Gmail or Outlook credential.
30. Set To field from `lead_email`.
31. Set Subject based on classification.
32. Use AI-generated or template-based email body.
33. Execute and verify email delivery.

Email Template Example

Dear {{\${json.lead_name}},

Thank you for your interest in AI automation for insurance operations.
Based on your requirement, our team can help you explore suitable automation options for {{\${json.insurance_use_case}}.

Our team will review your request and connect with you shortly.

Regards,
Insurance Automation Team

Mini Assignment 4

Create different email templates for Hot, Warm and Cold leads. Keep the tone professional and avoid overpromising business outcomes.

23. Hands-on Lab 7: Add Sales Notification

Objective: Notify sales team when a Hot Lead is received.

34. On the Hot Lead branch, add email, Slack, Teams or WhatsApp-approved notification node.
35. Include lead name, company, use case, urgency and recommended action.
36. Add CRM link if available.
37. Test with a Hot Lead sample.

Notification Example

```
Hot insurance automation lead received.  
Company: {{$json.company}}  
Use Case: {{$json.insurance_use_case}}  
Timeline: {{$json.timeline}}  
Recommended Action: {{$json.recommended_action}}
```

24. Hands-on Lab 8: Execution Review and Troubleshooting

Objective: Learn how to debug workflow execution in n8n.

- Use node execution view to inspect input and output data.
- Pin sample data during workflow design.
- Check credentials if API or email nodes fail.
- Validate JSON output from AI before sending to CRM.
- Add error branch for failed CRM push or email delivery.
- Keep sample test cases for Hot, Warm and Cold leads.

Common Mistakes

Mistake	Fix
Webhook receives empty data	Check method, URL and JSON body
AI output is plain text, not JSON	Update prompt: Return valid JSON only
CRM node fails	Check API URL, auth and required fields
Email not sent	Check SMTP/Gmail credentials and To field
IF branch not working	Check exact classification value and expression

25. Governance and Enterprise Best Practices

- Store all credentials in n8n credentials or approved secret storage.
- Avoid putting API keys directly inside prompts or workflow nodes.
- Log key business outputs but avoid storing unnecessary sensitive data.
- Validate AI outputs before updating CRM records.
- Use human review for high-value or sensitive leads.
- Create naming standards for workflows and nodes.
- Export workflow JSON for backup and version control where appropriate.
- Document owner, purpose, inputs, outputs and failure handling.

26. Small Assignments Summary

Assignment	Task	Expected Outcome
1	Create multiple lead samples	Webhook tested with varied data
2	Add lead_region field	Basic enrichment logic added
3	Improve AI prompt	Solution area extracted
4	Add follow-up date logic	CRM-ready next action field
5	Create three email templates	Hot, Warm and Cold response paths

27. Final Assignment: End-to-End n8n Automation Workflow

Build a complete lead processing workflow for an insurance automation business scenario.

Required Workflow

```
Webhook Lead Capture
-> Clean Lead Data
-> AI Lead Classification
-> IF Lead Priority
-> Push to CRM
-> Send Email Response
-> Notify Sales for Hot Leads
-> Log Execution
```

Required Output Fields

- lead_name
- lead_email
- company
- requirement
- classification
- priority
- insurance_use_case
- recommended_action
- email_response_status
- crm_push_status

Final Deliverables

- Working n8n workflow.
- Exported workflow JSON file.
- Three test executions: Hot Lead, Warm Lead and Cold Lead.
- Screenshot of successful CRM or spreadsheet entry.
- Screenshot or confirmation of automated email response.
- Short explanation of workflow design and business value.

28. Review Questions

38. What is the difference between a trigger and a node?

39. When should you use Webhook Trigger instead of Schedule Trigger?
40. Why should AI output be returned as JSON?
41. How can n8n connect with a CRM without custom code?
42. What are the risks of using AI classification without validation?
43. Where should credentials be stored?
44. What should be logged for enterprise audit?
45. How can UiPath and n8n work together in enterprise automation?

29. Day 6 Summary

Participants have learned how to use n8n as an enterprise orchestration layer for no-code and low-code automation. They built a practical marketing lead qualification workflow that captures lead data, classifies it with AI, routes it by priority, stores it in CRM and sends an automated response.

- n8n architecture and workflow concepts.
- Webhook, schedule and email trigger usage.
- Node-based workflow design.
- API integration without code.
- AI-powered lead classification.
- CRM push and automated email response.
- Enterprise governance and troubleshooting practices.

30. References

- n8n Docs: workflow automation documentation and built-in node guides.
- n8n Webhook node documentation: using Webhook node to trigger workflows.
- n8n Schedule Trigger node documentation: running workflows on fixed schedules.
- n8n AI workflow tutorial: building AI workflows with n8n.